

Pilot and Trial Search Strings (Preliminary Search Iterations)

Before defining the final search strings used in the systematic literature review, several **pilot and trial search queries** were formulated and tested. These preliminary strings were used to (i) validate keyword coverage, (ii) assess recall and precision, and (iii) iteratively refine terminology aligned with the research questions. The pilot searches were executed on IEEE Xplore, Scopus, and ACM Digital Library.

Pilot String 1: Core Three-Area Intersection

This initial trial explicitly targeted all three areas using only canonical terminology.

("large language model" OR LLM OR ChatGPT OR GPT)
AND
("software architecture" OR "architectural design")
AND
("self-adaptive system" OR "self-adaptive software systems")

Observation

Returned very few results; many relevant studies used alternative terminology for self-adaptation or architectural activities.

Pilot String 2: Expanded Software Architecture Design Keywords

LLMs, Architectural design and reasoning terms were added to capture software architecture activities beyond high-level mentions.

("large language model" OR LLM OR "generative AI" OR ChatGPT OR GPT)
AND
("software architecture" OR "architecture design" OR
"architectural reasoning" OR "architecture evaluation")
AND
("self-adaptive system" OR "self-adaptive software systems" OR "SASS")

Observation

Improved architectural relevance, but several adaptive systems papers used different adaptation-related terminology.

Pilot String 4: Expanded Self-Adaptation and Autonomy Concepts

To reflect common terminology in SASS and CASA literature, all terms were generalized using wildcards.

("large language model" OR LLM OR "generative AI" OR GPT OR ChatGPT OR "Transformer")

AND

("software architecture" OR "architecture design" OR "architecture reasoning" OR "architecture evaluation" OR "architectural modeling")

AND

(adapt* OR self*)

Pilot String 5: Generalized LLMs

("large language model" OR "large language models" OR LLM OR LLMs OR "generative AI" OR "generative Artificial Intelligence" OR "Gen AI" OR "Gen Artificial Intelligence" OR "generative model" OR "Transformer model" OR "Transformer models") AND ("software architecture" OR "architecture design" OR "architectural design" OR "architecture modeling" OR "architectural modeling" OR "architecture evaluation" OR "architecture analysis" OR "architectural reasoning" OR "architecture decision-making" OR "design rationale" OR "decision support" OR "architecture framework" OR "architecture pattern" OR "architecture tactic" OR "trade-off analysis" OR "quality attribute analysis" OR "design decision") AND (adapt* OR self* OR autonom*)

Pilot String 6: Generalized LLMs

("large language model" OR "large language models" OR LLM OR LLMs OR "generative AI" OR "generative Artificial Intelligence" OR "Gen AI" OR "Gen Artificial Intelligence") AND ("software architecture" OR "architecture design" OR "architectural design" OR "architecture modeling" OR "architectural modeling" OR "architecture evaluation" OR "architecture analysis" OR "architectural reasoning" OR "architecture decision-making" OR "design rationale" OR "decision support" OR "architecture framework" OR "architecture pattern" OR "architecture

tactic" OR "trade-off analysis" OR "quality attribute analysis" OR "design decision") AND (adapt* OR self* OR autonom*)

Note

The pilot search strings were exploratory and used solely to refine keyword coverage and terminology. The final search string represents the stabilized version aligned with the research questions and was systematically adapted to each digital library.

Final Search String used for Review (2018-2025)

("large language model" OR "large language models" OR LLM OR LLMs OR "generative AI" OR "generative Artificial Intelligence" OR "Gen AI" OR "Gen Artificial Intelligence") AND ("software architecture" OR "architecture design" OR "architectural design" OR "architecture modeling" OR "architectural modeling" OR "architecture evaluation" OR "architecture analysis" OR "architectural reasoning" OR "architecture decision-making" OR "design rationale" OR "decision support" OR "architecture framework" OR "architecture pattern" OR "architecture tactic" OR "trade-off analysis" OR "quality attribute analysis" OR "design decision") AND (adapt* OR self* OR autonom*)

Note: This is the general query that we generated by extracting keywords from our Research Questions.

IEEE Xplore

((("Document Title":"large language model" OR "Document Title":"large language models" OR "Document Title":LLM OR "Document Title":LLMs OR "Document Title":"generative AI" OR "Document Title":"generative Artificial Intelligence" OR "Document Title":"Gen AI" OR "Document Title":"Gen Artificial Intelligence" OR "Abstract":"large language model" OR "Abstract":"large language models" OR "Abstract":LLM OR "Abstract":LLMs OR "Abstract":"generative AI" OR "Abstract":"generative Artificial Intelligence" OR "Abstract":"Gen AI" OR "Abstract":"Gen Artificial Intelligence") AND ("Document Title":"software architecture" OR "Document Title":"architecture design" OR "Document Title":"architectural design" OR "Document Title":"architecture modeling" OR "Document Title":"architectural modeling" OR "Document Title":"architecture evaluation" OR "Document Title":"architecture analysis" OR "Document Title":"architectural reasoning" OR "Document Title":"architecture decision-making" OR "Document Title":"design rationale" OR "Document Title":"decision support" OR "Document Title":"architecture framework" OR "Document Title":"architecture pattern" OR "Document

Title:"architecture tactic" OR "Document Title":"trade-off analysis" OR "Document Title":"quality attribute analysis" OR "Document Title":"design decision" OR "Abstract":"software architecture" OR "Abstract":"architecture design" OR "Abstract":"architectural design" OR "Abstract":"architecture modeling" OR "Abstract":"architectural modeling" OR "Abstract":"architecture evaluation" OR "Abstract":"architecture analysis" OR "Abstract":"architectural reasoning" OR "Abstract":"architecture decision-making" OR "Abstract":"design rationale" OR "Abstract":"decision support" OR "Abstract":"architecture framework" OR "Abstract":"architecture pattern" OR "Abstract":"architecture tactic" OR "Abstract":"trade-off analysis" OR "Abstract":"quality attribute analysis" OR "Abstract":"design decision") AND ("Document Title":adapt* OR "Document Title":self* OR "Document Title":autonom* OR "Abstract":adapt* OR "Abstract":self* OR "Abstract":autonom*))

Note: The studies retrieved using the IEEE Xplore search string can be found in a worksheet named “IEEEXplore_Analysis” inside the file "Master_Index.xlsx" the replication package.

Scopus

TITLE-ABS("large language model" OR "large language models" OR "LLM" OR "LLMs" OR "generative AI" OR "generative Artificial Intelligence" OR "Gen AI" OR "Gen Artificial Intelligence") AND TITLE-ABS("software architecture" OR "architecture design" OR "architectural design" OR "architecture modeling" OR "architectural modeling" OR "architecture evaluation" OR "architecture analysis" OR "architectural reasoning" OR "architecture decision-making" OR "design rationale" OR "decision support" OR "architecture framework" OR "architecture pattern" OR "architecture tactic" OR "trade-off analysis" OR "quality attribute analysis" OR "design decision") AND TITLE-ABS(adapt* OR self* OR autonom*)

Note: The studies retrieved using the IEEE Xplore search string can be found in a worksheet named “Scopus_Analysis” inside the file "Master_Index.xlsx" inside the replication package.

ACM/DL

((Title:("large language model" OR "large language models" OR "LLM" OR "LLMs" OR "generative AI" OR "generative Artificial Intelligence" OR "Gen AI" OR "Gen Artificial Intelligence") OR Abstract:("large language model" OR "large language models" OR "LLM" OR "LLMs" OR "generative AI" OR "generative Artificial Intelligence" OR "Gen AI" OR "Gen Artificial Intelligence"))) AND (Title:("software architecture" OR "architecture design" OR "architectural design" OR "architecture modeling" OR "architectural modeling" OR "architecture evaluation" OR "architecture analysis" OR "architectural reasoning" OR "architecture decision-making" OR "design rationale" OR "decision support" OR "architecture framework" OR

"architecture pattern" OR "architecture tactic" OR "trade-off analysis" OR "quality attribute analysis" OR "design decision") OR Abstract:("software architecture" OR "architecture design" OR "architectural design" OR "architecture modeling" OR "architectural modeling" OR "architecture evaluation" OR "architecture analysis" OR "architectural reasoning" OR "architecture decision-making" OR "design rationale" OR "decision support" OR "architecture framework" OR "architecture pattern" OR "architecture tactic" OR "trade-off analysis" OR "quality attribute analysis" OR "design decision")) AND (Title:(adapt* OR self* OR autonom*) OR Abstract:(adapt* OR self* OR autonom*)))

Note: The studies retrieved using the IEEE Xplore search string can be found in a worksheet named “ACMDL_Analysis” inside the file "Master_Index.xlsx" inside the replication package.